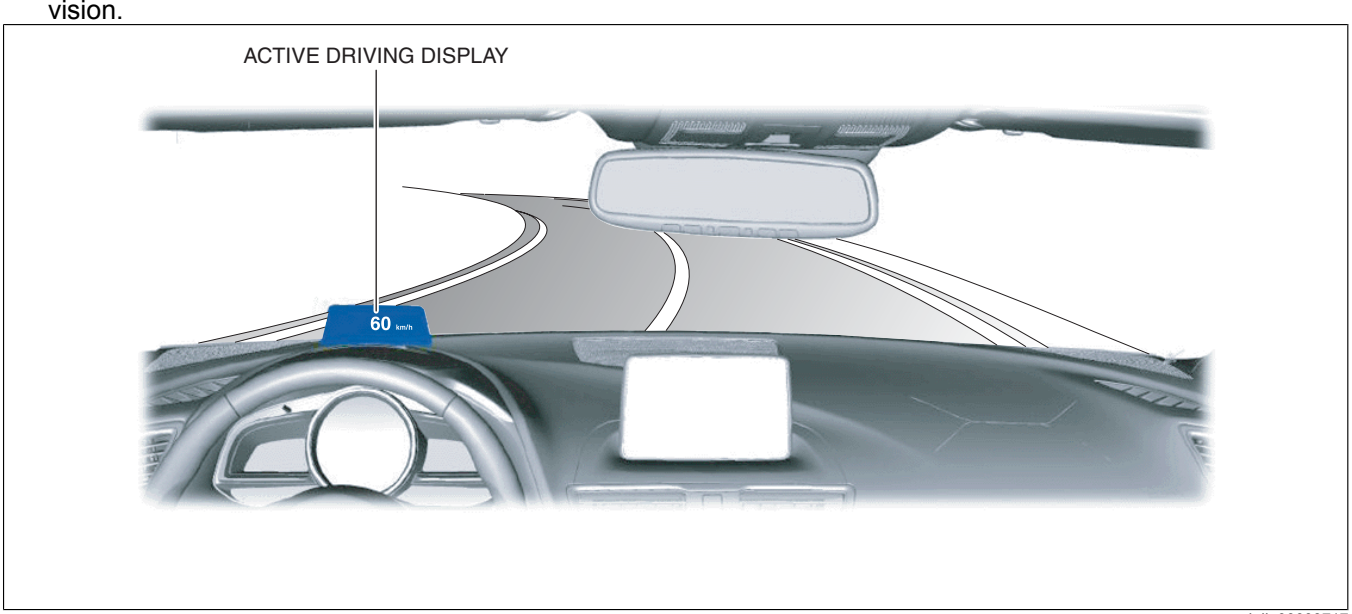
ACTIVE DRIVING DISPLAY

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Purpose

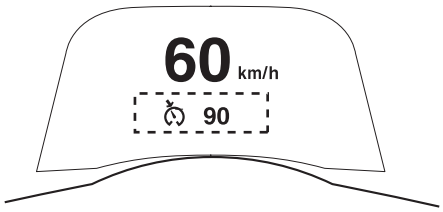
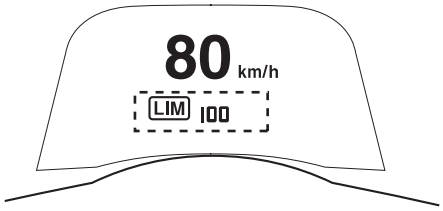
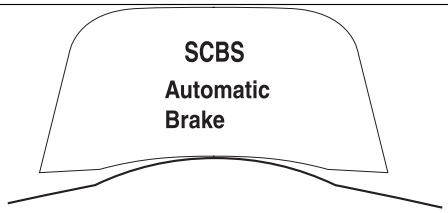
- The active driving display displays the vehicle information in the display which is set within the forward visual range of the driver, allowing verification of the vehicle information with only the slightest change in the line of vision.

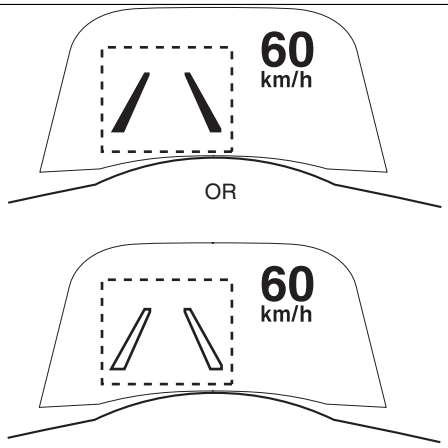
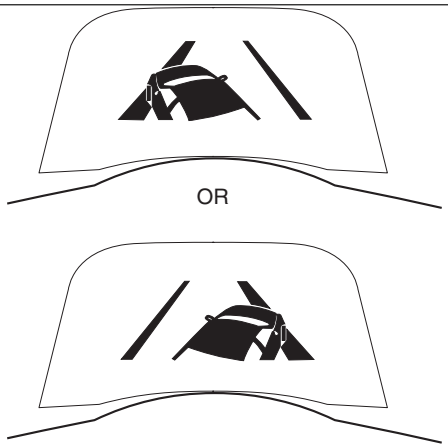
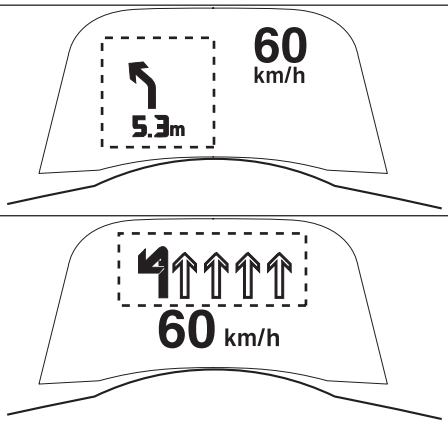


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Function

- The active driving display displays the following vehicle information.
  - — Current vehicle speed
  - — Cruise control set vehicle speed (With cruise control system)
  - — Adjustable speed limiter set vehicle speed (With adjustable speed limiter)
  - — Smart city brake support (SCBS) warning display (With smart city brake support (SCBS))
  - — Lane departure warning system (LDWS) operation status or warning display (With lane departure warning system (LDWS))
  - — Travel direction (turn by turn (TBT)) display (With car navigation system)
- When the engine is started, the instrument cluster displays the following content on the active driving display based on the following CAN signals. For details on each display content, refer to each reference.

CAN signal sending module	Signal name	Display content	Display example	Reference
PCM	Vehicle speed signal	Current vehicle speed		(See SPEEDOMETER.)
	Cruise control set vehicle speed display signal	Cruise control set vehicle speed		(See CRUISE CONTROL SYSTEM [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CRUISE CONTROL SYSTEM [SKYACTIV-D 1.5].)
	Adjustable speed limiter set vehicle speed display signal	Adjustable speed limiter set vehicle speed		(See ADJUSTABLE SPEED LIMITER [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ADJUSTABLE SPEED LIMITER [SKYACTIV-D 1.5].)
Laser sensor	Warning display request signal	Warning screen		(See SMART CITY BRAKE SUPPORT (SCBS).)

CAN signal sending module	Signal name	Display content	Display example	Reference
Forward sensing camera (FSC)	Operation display request signal	Vehicle lane display		(See LANE DEPARTURE WARNING SYSTEM (LDWS).)
	Warning display request signal	Warning screen		
Connectivity master unit (CMU)	Turn-by-turn (TBT) display request signal	Travel direction display		(See ENTERTAINMENT SYSTEM [WITH CENTER DISPLAY].)

Setting function

- The following items for the active driving display can be set on the [Settings] screen of the center display.

Item	Setting contents
Height	The display open/close angle can be adjusted.
Brightness Control	Adjustment of the display brightness can be switched between automatic and manual.
Calibration	The display's automatic dimness adjustment level can be switched. (Brightness adjustment must be set to automatic.)
Brightness	The brightness of the display can be adjusted. (Brightness adjustment must be set to manual.)
Navigation	The turn-by-turn (TBT) display can be switched on/off.
Active Driving Display	Display/non-display of the active driving display can be switched.
Reset	The active driving display settings can be reset to the initial settings.

Demonstration function

Note
- The demonstration function force-displays the active driving display without the engine being started, such as on an indoor exhibition vehicle, so that customers can verify the display status.

- The demonstration function launches the active driving display with the ignition switched ON (engine off) to display the vehicle information in the display.

Demonstration function on/off setting procedure

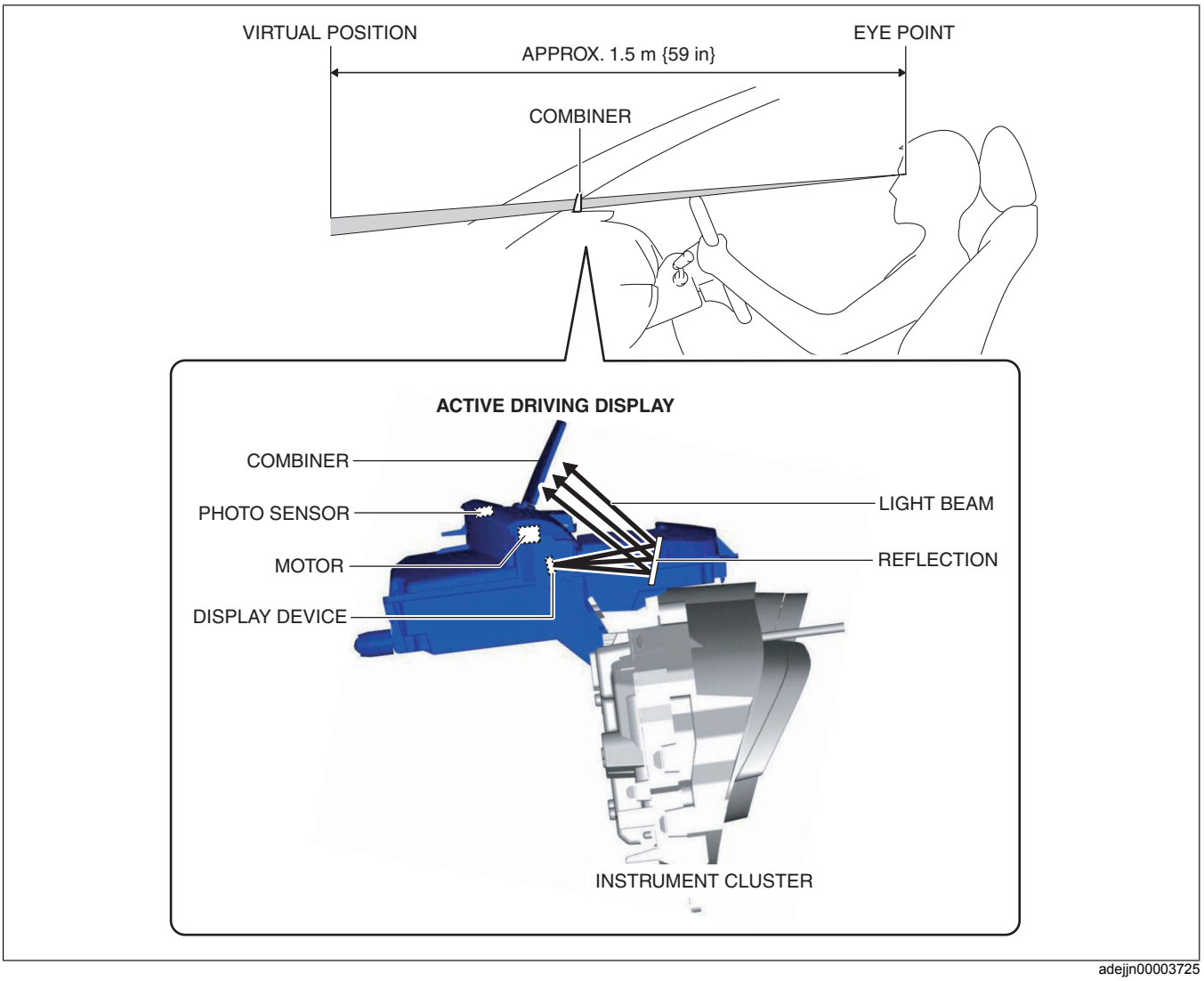
1. With the tripmeter switch in the instrument cluster pressed, switch the ignition as follows.
   - Ignition off→Ignition ON (engine off)→Ignition off→Ignition ON (engine off)
2. Release the tripmeter switch when [test] is displayed in the instrument cluster.

Note
- When the tripmeter switch is released with [test] displayed in the instrument cluster, the entire instrument cluster is turned on.

3. Press the tripmeter switch and display [06 on] or [06 off] in the instrument cluster.
   - Select [06 on] to turn on the demonstration function.
   - Select [06 off] to turn off the demonstration function.
4. Switch the ignition off.

Note
- If the ignition is switched off with [06 on] displayed in the instrument cluster (demonstration function on), the demonstration function will operate the next time the ignition is switched ON (engine off).
- If the following is performed with the demonstration function on, the demonstration function turns off.
  — The battery power is interrupted or the ROOM fuse is removed.
  — The vehicle is driven at **20 km/h {12 mph} or more**.

Construction
- The active driving display is assembled to the instrument cluster.
- The active driving display consists of a display device which produces the display content and emits the content over a light beam, a reflector which reflects the light beam emitted from the display device, a combiner which reflects the light beam from the reflector and displays the screen, a motor which opens/closes the combiner, and a photo sensor which detects the brightness surrounding the display device.
- The combiner is a semi-transparent acrylic board processed for reflecting the light beam from the display device.
- The combiner opens at the top of the meter hood and displays vehicle information in the forward visual range of the driver. As a result, the time has been shortened in which the driver's line of vision changes from the view at the front of the vehicle to the view of the vehicle information reflected in the combiner.
- By using a concave shaped display, the image reflected in the display is positioned virtually **approx. 1.5 m {59 in} forward** of the eye point. As a result, the eye point adjustment time has been reduced in which the driver's line of vision changes from the view at the front of the vehicle to the view of the vehicle information reflected in the display.



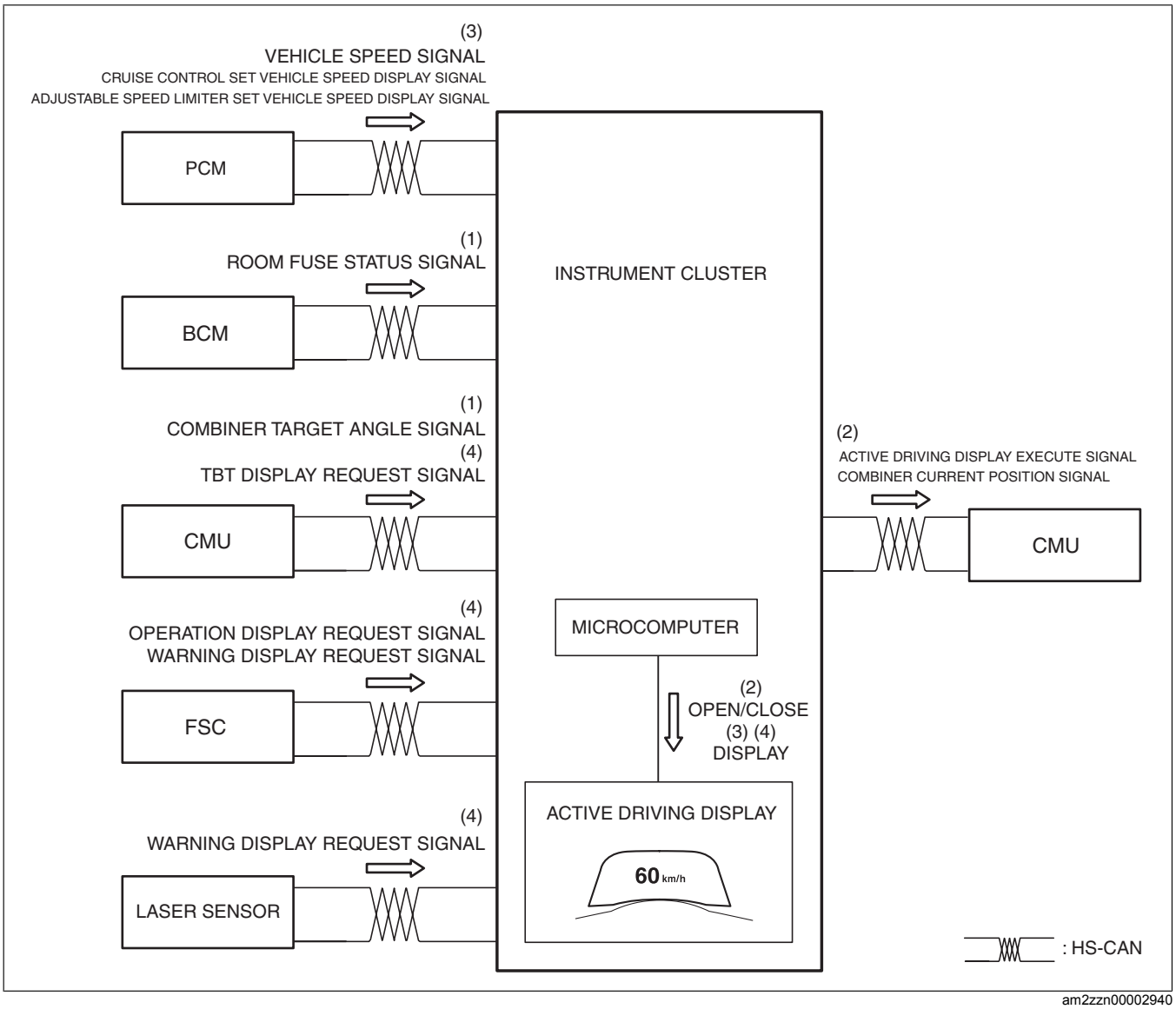
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Operation

1. When the ignition is switched ON (engine on), the instrument cluster opens the combiner (1) based on the following CAN signals.
   — ROOM fuse status signal from BCM
   — Combiner target angle signal from connectivity master unit (CMU)
2. When the combiner opens, the instrument cluster displays the opening screen (vehicle name) for **approx. 2 s**. In addition, it sends an active driving display execute signal and combiner current position signal (2) to the connectivity master unit (CMU).

   Note
   - The connectivity master unit (CMU) records the current position of the combiner based on the combiner current position signal from the instrument cluster. When the ignition is switched from off to ON (engine on), it sends the target angle signal based on the recorded current position.
   - The combiner current position recorded by the connectivity master unit (CMU) is not cleared even if the battery power is disconnected.
   - If the connectivity master unit (CMU) has not recorded the combiner current position, it sends the target angle signal of the initial value.

3. After the opening screen is turned off, the instrument cluster displays the current vehicle speed based on the vehicle speed signal (3) from the PCM, and displays the warning or other screens based on request signals (4) from related modules.



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Fail-safe

- If the instrument cluster detects a malfunction in the combiner drive motor, it turns off the active driving display images and stops the display at the current position.